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L_RAG: Knowledge-Graph-Augmented Hybrid Legal RAG System

System Architecture, Corpus Design, and Retrieval Components

Course: Text Mining

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Abstract

Legal Information Retrieval (IR) and Question Answering (QA) pose significant technical challenges due to domain-specific terminology, complex document hierarchies, temporal validity constraints, and cross-document statutory references. Standard dense retrieval systems often fail to capture exact statutory citations and cross-article dependencies, while traditional keyword search misses semantic paraphrasing. In this work, we introduce **L_RAG**, a comprehensive Knowledge-Graph-augmented hybrid RAG architecture designed for Vietnamese legal corpora. The system integrates dual-stage retrieval (Dense **multilingual-e5-large** + Sparse BM25), Knowledge Graph dynamic overlays for legal validity and structural traversal, Reciprocal Rank Fusion (RRF), multi-reranker candidate refinement (CrossEncoder, BGE-v2, Jina-v2, Qwen3-Reranker), and citation-grounded LLM synthesis. This document presents a complete description of the system architecture, component specifications, knowledge graph schema, and corpus design up to the experimental evaluation setup.

1 Introduction

Legal Information Retrieval (IR) and Question Answering (QA) represent critical components in modern legal technology, enabling legal professionals, citizens, and researchers to efficiently navigate vast repositories of statutes, decrees, circulars, and judicial precedents. However, applying general-purpose Retrieval-Augmented Generation (RAG) models to legal domains presents distinct domain-specific challenges. Unlike open-domain Wikipedia-style QA, legal reasoning demands strict grounding in authoritative statutory context, precise citation of applicable legal provisions (e.g., Articles, Clauses, Items), and strict temporal validity verification (ensuring that superseded or annulled laws are not cited).

1.1 Problem Statement

Conventional RAG pipelines rely heavily on single-stage dense vector retrieval powered by pre-trained embedding models. While dense retrieval captures high-level semantic intent, it suffers from three core limitations when deployed on complex Vietnamese legal corpora:

- 1. The Semantic-Statutory Vocabulary Gap:** Natural language user queries (e.g., *"Tôi bị phạt bao nhiêu tiền khi không đội mũ bảo hiểm?"*) differ significantly in terminology from legal codifications (e.g., *"Điều 6 Nghị định 100/2019/NĐ-CP: Phạt tiền từ 300.000 đồng đến 400.000 đồng..."*). Pure vector search often maps natural queries to general descriptions rather than the exact punitive provision.
- 2. Structural and Hierarchy Blindness:** Legal documents follow strict hierarchical compositions (Law $\rightarrow$ Chapter $\rightarrow$ Article $\rightarrow$ Clause $\rightarrow$ Item). Standard fixed-length text chunking fragments these structural relationships, causing vector search to retrieve isolated clauses missing their governing article definitions or parent law scope.
- 3. Temporal and Inter-Document Citation Complexity:** Legal provisions do not exist in isolation; they amend, replace, repeal, or reference other statutory provisions across multiple legal instruments. Pure semantic similarity fails to account for statutory precedence (e.g., Constitution $\succ$ Law $\succ$ Decree) or dynamic validity timelines.

To resolve these challenges, a legal RAG system must unify dense semantic matching, exact sparse keyword matching, statutory structure traversal, and fine-grained cross-encoder reranking.

1.2 Objectives

The primary goal of the **L_RAG** project is to design, implement, and analyze a novel, robust Knowledge-Graph-Augmented Hybrid RAG framework specifically engineered for Vietnamese legal corpora. The specific technical objectives are:

- **Architect an End-to-End Hybrid RAG Pipeline:** Combine dense vector retrieval (`intfloat/multilingual-e5-large`), sparse tokenized keyword search (`BM25`), and Reciprocal Rank Fusion (RRF) to maximize recall across diverse query types.
- **Develop a Knowledge Graph (G-LRAG) Overlay Module:** Construct an in-memory legal knowledge graph that models containment hierarchies (`DOCUMENT_HAS_PROVISION`, `PROVISION_HAS_CHUNK`) and structural reading orders (`CHUNK_NEXT`, `PROVISION_NEXT`) to dynamically supply document filtering whitelists and same-provision context expansions.
- **Integrate Advanced Multi-Reranker Pipelines:** Deploy and evaluate local Cross-Encoder models—including `mMiniLMv2`, `BGE-Reranker-v2-m3`, `Jina-Reranker-v2`, and `Qwen3-Reranker-0.6B`—to refine candidate rankings prior to LLM generation.
- **Construct a Standardized 500-Query Vietnamese Legal Benchmark:** Curate a comprehensive benchmark of 500 legal QA pairs covering answerable, unanswerable, single-hop, multi-hop, citation, cross-document, and legal validity categories.

1.3 Contributions

The key contributions of this work include:

1. **Comprehensive System Implementation:** We deliver a production-ready, modular codebase supporting end-to-end legal data ingestion, vector indexing, knowledge graph building, hybrid fusion, reranking, and citation-grounded LLM synthesis (`gpt-4o-mini`).
2. **Dynamic Knowledge Graph Validity & Traversal Engine:** We implement query-time overlay joining and precedence sorting algorithms that enforce legal hierarchy and temporal validity without requiring rigid database re-indexing.
3. **Multi-Reranker Benchmarking & Optimization:** We provide an extensive empirical comparison across four state-of-the-art cross-encoders, introducing optimized FP16 batching and monkey-patch solutions for seamless Kaggle GPU execution.

1.4 Report Structure

The remainder of this report is organized as follows: Section 2 reviews related work in RAG, Legal IR, hybrid search, and Knowledge-Graph-augmented systems. Section 3 presents the overall L_RAG system architecture and describes each component (retrieval, knowledge graph, reranking, generation) in detail. Section 4 details the Vietnamese legal corpus, knowledge graph schema, and the 500-query benchmark dataset. Section ?? through Section ?? (in subsequent chapters) define the experimental protocol, present ablation results, and analyze system trade-offs.

2 Related Work

This section reviews existing literature and foundational paradigms relevant to Retrieval-Augmented Generation, domain-specific Legal Information Retrieval, hybrid retrieval strategies, and Knowledge-Graph-augmented systems.

2.1 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation [?] has emerged as a dominant framework to mitigate hallucinations, overcome outdated parameter knowledge, and ground Large Language Models (LLMs) in verifiable external knowledge. Standard RAG architectures operate in two main steps: (1) retrieving relevant document chunks from a vector store using dense bi-encoder embeddings, and (2) concatenating retrieved chunks into the prompt context for an LLM to generate an answer.

Despite their success in general-domain benchmarks, baseline RAG architectures face severe limitations in specialized, highly technical domains. In unstructured domain corpora, dense retrieval frequently suffers from low precision due to semantic ambiguity, while fixed-size text chunking loses structural dependencies spanning across paragraphs or document chapters. Advanced RAG paradigms—such as Self-RAG [?] and Corrective RAG (CRAG) [?—introduce self-reflection and web-search fallback mechanisms, but they do not explicitly model formal domain taxonomies or statutory legal validity.

2.2 Legal Information Retrieval

Legal Information Retrieval (Legal IR) is a specialized subfield at the intersection of Information Retrieval and Legal Informatics [?]. Legal IR differs fundamentally from open-domain text search due to four intrinsic characteristics of legal text:

1. **Hierarchical Structure:** Legal statutes are organized into strict logical trees (Code → Part → Chapter → Section → Article → Clause → Item).
2. **High Precision Requirements:** A single misplaced word or omission of an exception clause can alter the legal interpretation completely.
3. **Statutory Precedence:** Legal systems follow formal legal hierarchies (e.g., the Constitution overrides National Assembly Laws, which in turn override Government Decrees and Ministry Circulars).
4. **Dynamic Validity & Amendments:** Legal provisions are constantly updated, amended, or repealed by subsequent legal instruments over time.

Early Legal IR research focused on rule-based citation parsing and expert systems. Recent approaches leverage specialized transformer models, such as Legal-BERT [?] and Law-Former [?], to encode statutory texts. However, relying solely on neural embeddings remains insufficient for capturing complex statutory relationships and temporal applicability.

2.3 Hybrid Retrieval and Reranking Fusion

To overcome the vocabulary mismatch problem of dense embeddings and the semantic rigidity of sparse keyword search, hybrid retrieval combines both modalities [?]. Sparse algorithms (e.g., BM25) excel at exact keyword matching—critical for capturing specific legal codes, numbers, and technical terminology—whereas dense bi-encoders (e.g., E5 [?]) excel at conceptual paraphrasing.

Combining retrieved lists from heterogeneous engines requires robust rank fusion. Reciprocal Rank Fusion (RRF) [?] merges ranked outputs based on positional reciprocal scores:

$$\text{RRF_Score}(d \in D) = \sum_{m \in M} \frac{1}{k + r_m(d)} \quad (1)$$

where M is the set of retrieval systems, $r_m(d)$ is the rank of document d in system m , and k is a smoothing constant (typically $k = 60$).

Following initial retrieval, cross-encoder rerankers [?] evaluate full joint attention between the query and candidate passages (q, d) . Unlike bi-encoders, which compute separate vectors, cross-encoders achieve significantly higher ranking precision (measured via MRR and nDCG) at the expense of higher inference latency. Recent state-of-the-art multilingual rerankers include BGE-Reranker-v2-m3 [?], Jina-Reranker-v2 [?], and Qwen3-Reranker [?].

2.4 Knowledge-Graph-Augmented Retrieval (GraphRAG)

Knowledge Graphs (KGs) represent domain entities and their explicit relationships as structured nodes and edges (e_1, r, e_2) . GraphRAG [?] integrates graph traversal with vector search, enabling LLMs to reason over multi-hop relationships and hierarchical structures. In the legal domain, GraphRAG approaches construct statutory graphs linking documents, provisions, definitions, and legal concepts. At query time, graph traversal algorithms (e.g., Breadth-First Search) expand seed vector hits by navigating structural edges (`HAS_PROVISION`) and citation edges (`CITES`, `AMENDS`). This guarantees that retrieved contexts maintain statutory completeness and legal coherence.

3 System Overview: L_RAG

This section details the end-to-end architecture of **L_RAG**, a Knowledge-Graph-augmented hybrid RAG system engineered for legal information retrieval and question answering over Vietnamese legal corpora.

3.1 Architecture

The L_RAG pipeline unifies structured graph processing, hybrid vector-sparse search, cross-encoder candidate reranking, and citation-grounded LLM synthesis. Figure 1 illustrates the system dataflow across four distinct operational phases: (1) Corpus Ingestion & Graph/Vector Indexing, (2) Multi-Modal Hybrid Retrieval, (3) Knowledge Graph Overlay & Context Expansion, (4) Sequential Candidate Reranking, and (5) LLM Prompting & Generation.

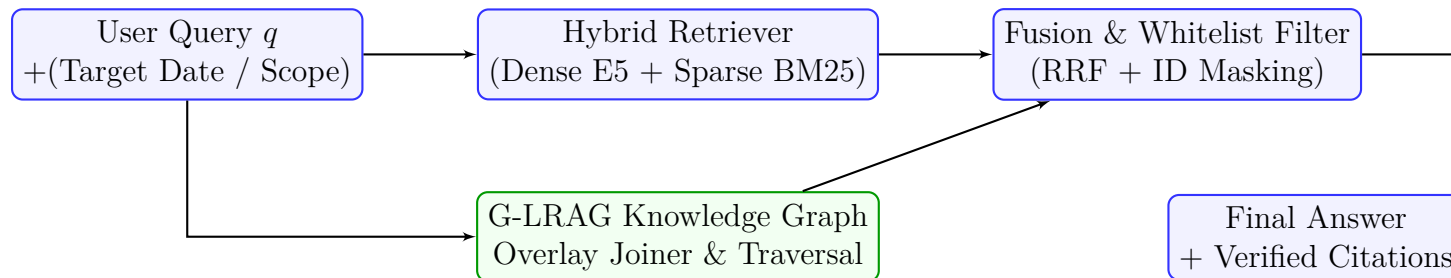


Figure 1: End-to-end operational workflow of the L_RAG system.

3.2 Retrieval Components

The retrieval module combines three complementary retrieval approaches:

3.2.1 Dense Vector Search

Dense retrieval is powered by `intfloat/multilingual-e5-large`, a 560M parameter transformer embedding model. To solve the structural blindness of standalone chunks, each text chunk is preprocessed into a **Metadata-Enriched Representation**:

$$\text{Text}_{\text{embedded}} = \text{Doc_Title} \parallel \text{Provision_Header} \parallel \text{Chunk_Content} \quad (2)$$

Chunks are stored in an optimized FAISS vector database using `IndexFlatIP` with L2-normalized embeddings for fast cosine similarity search.

3.2.2 Sparse Keyword Search (BM25)

Sparse retrieval utilizes an inverted index built with `rank_bm25`. Raw legal texts are tokenized using the `underthesea` Vietnamese morphological tokenizer, ensuring compound legal terms (e.g., *"bắt động sản"*, *"trách nhiệm hình sự"*) are indexed as unified lexical tokens. BM25 parameters are set to standard defaults ($k_1 = 1.5, b = 0.75$).

3.2.3 Knowledge Graph Overlay (G-LRAG Module)

The Knowledge Graph module acts as a query coordinator that generates dynamic document filter whitelists (`id_str_filter`) and performs horizontal reading-order expansions. Key features include:

- **Dynamic Overlay Folding:** Validity timelines are joined at query time relative to an `as_of_date` parameter, ensuring superseded provisions are excluded.
- **Precedence Sorting:** Conflicts between legal authority ranks are resolved using statutory hierarchy rules:

$$\text{Hiến pháp} \succ \text{Bộ luật} / \text{Luật} \succ \text{Nghị định} \succ \text{Thông tư} \quad (3)$$

- **Reading-Order Traversal:** Structural edge pointers (`CHUNK_NEXT`, `PROVISION_NEXT`) slide context windows horizontally across sibling chunks within the same parent provision.

3.2.4 Fusion Layer

Search hit lists from dense search (R_{dense}) and sparse search (R_{BM25}) are merged using Reciprocal Rank Fusion (RRF) with constant $k = 60$. In configurations without RRF, score normalization is applied using min-max scaling prior to rank merging.

3.3 Reranking Components

Initial retrieval selects $K = 30$ candidate chunks per query. To maximize precision, candidate lists pass through a specialized Cross-Encoder Reranker that scores the full $(query, chunk_text)$ pair.

L_RAG supports four distinct reranking backends loaded sequentially onto GPU:

1. **mMiniLMv2: cross-encoder/mmarco-mMiniLMv2-L12-H384-v1** (Lightweight baseline, 117M params).
2. **BGE-Reranker-v2-m3**: BAAI/bge-reranker-v2-m3 (Multilingual 567M params with sigmoid activation).
3. **Jina-Reranker-v2**: jinaai/jina-reranker-v2-base-multilingual (278M params, featuring dynamic position embedding patches).
4. **Qwen3-Reranker**: Qwen/Qwen3-Reranker-0.6B (CausalLM-based 0.6B parameter model evaluated in FP16 precision).

To prevent GPU Out-Of-Memory (OOM) failures during multi-model evaluation, the loader executes a strict **Sequential Load-Run-Unload Cycle**:

$$\text{Load}(M_i) \longrightarrow \text{Execute}(M_i) \longrightarrow \text{Delete}(M_i) \longrightarrow \text{torch.cuda.empty_cache()} \quad (4)$$

3.4 Generation Component

The top $N = 10$ reranked chunks are formatted into a structured prompt context supplied to **gpt-4o-mini** (temperature $T = 0.0$, max tokens 1024).

The system prompt enforces three critical compliance rules:

1. **Strict Citation Grounding**: Answers must explicitly cite governing statutory sources (e.g., "*Theo Điều 15, Luật Đất đai 2024...*").
2. **Unanswerable Abstention**: If the retrieved context lacks sufficient evidence, the LLM must output `INSUFFICIENT_CONTEXT` rather than hallucination.
3. **Answer-Type Formatting**: Outputs are formatted according to the query type (*boolean*, *extractive*, *abstractive*, or *unanswerable*).

4 Dataset and Benchmark

This section describes the Vietnamese legal corpus, the structural design of the G-LRAG Knowledge Graph, and the 500-query legal QA benchmark dataset used to evaluate the L_RAG system.

4.1 Corpus Description

The legal dataset comprises a massive collection of Vietnamese legal instruments, including National Assembly Laws, Government Decrees, Prime Minister Decisions, and Ministry Circulars. Raw legal documents undergo structured parsing to preserve document metadata, hierarchical provisions, and passage chunks.

Table 1 summarizes the core scale and statistics of the processed corpus.

Table 1: Overall Statistics of the Vietnamese Legal Corpus.

Entity Type	Description	Total Count
Documents	Full legal instruments (Laws, Decrees, etc.)	151,624
Provisions	Hierarchical units (Articles, Clauses, Items)	1,386,267
Chunks	Fixed/Structural text passages for retrieval	1,513,376
Document Edges	Relationships (Amend, Replace, Cite, Repeal)	883,256
External Stubs	Referenced external entities	19,763

Each passage chunk is enriched with metadata prior to indexing:

$$\text{Chunk_Meta} = \{\text{chunk_id}, \text{doc_id}, \text{provision_id}, \text{title}, \text{issuing_authority}, \text{effective_date}\} \quad (5)$$

4.2 Knowledge Graph Structure (G-LRAG Schema)

The G-LRAG Knowledge Graph models statutory entities as nodes and structural/legal dependencies as typed directed edges.

4.2.1 Node Schema

- **Document Node (D):** Represents a full legal instrument. Attributes: `doc_id`, `title`, `authority_rank`, `issuing_body`, `effective_date`, `validity_status`.
- **Provision Node (P):** Represents a logical statutory section (Article/Clause). Attributes: `provision_id`, `header`, `provision_type`.

- **Chunk Node (C):** Represents an indexed passage. Attributes: `chunk_id`, `text`, `sequence_num`.

4.2.2 Edge Schema

Relationships are divided into structural hierarchy and inter-document legal relationships:

- **Structural Hierarchy Edges:**
 - `DOCUMENT_HAS_PROVISION`: Connects Document D to child Provision P .
 - `PROVISION_HAS_CHUNK`: Connects Provision P to child Chunk C .
 - `CHUNK_NEXT`: Sequentially links adjacent sibling chunks $C_i \rightarrow C_{i+1}$ within reading order.
 - `PROVISION_NEXT`: Sequentially links adjacent provisions $P_j \rightarrow P_{j+1}$.
- **Legal Relationship Edges:**
 - `AMENDS`: Indicates document D_A amends provision/document D_B .
 - `REPLACES`: Indicates document D_A replaces document D_B .
 - `CITES`: Indicates document D_A references document D_B .

4.2.3 Quarantine & Verification Integrity Gating

To prevent graph poisoning, G-LRAG enforces two strict integrity rules:

1. **Quarantine Gating:** Unverified or draft records are quarantined during ingestion.
2. **Verified-Only Traversal:** Path traversals only traverse edges where `direction_verified == True`.

4.3 QA Benchmark Dataset

To rigorously evaluate legal retrieval and generation quality, we curated a benchmark dataset of **500 Vietnamese legal questions** (`qa_final.jsonl`).

4.3.1 Dataset Composition

The benchmark balances real-world legal scenarios across multiple dimensions:

- **Answerability:** 400 *Answerable* questions (grounded in corpus evidence) and 100 *Unanswerable* questions (designed to test hallucination resistance).

- **Question Categories:**

1. *Citation*: Questions asking for specific article/law numbers.
2. *Cross-Document*: Questions requiring information synthesis across multiple legal documents.
3. *Legal Validity*: Questions regarding whether a regulation is currently effective.
4. *Multi-Hop*: Complex queries requiring step-by-step reasoning over connected provisions.
5. *Single-Hop*: Direct factual queries over a single provision.

- **Answer Types:** *Abstractive* (summary/explanation), *Boolean* (Yes/No), *Extractive* (verbatim quote), and *Unanswerable*.

4.4 Ground Truth Construction

Each answerable query is annotated by legal domain experts with gold-standard metadata:

$$Q_i = \{\text{qa_id}, \text{question}, \text{category}, \text{difficulty}, \text{answer_type}, \text{ref_answer}, \text{ground_truth_chunk_ids}\} \quad (6)$$

The `ground_truth_chunk_ids` set forms the exact retrieval baseline against which $\text{Recall}@k$, $\text{Hit}@k$, $\text{MRR}@k$, and $\text{nDCG}@k$ are evaluated.